

Conditional Expectation

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First, an example

Roll two fair six-sided dice. Let Y be the value of the first die and X be the sum of both dice. Given that the first die shows $Y = y$, the sum equals y plus the value of the second die, which has expectation 3.5 regardless of y . So

$$E[X \mid Y = y] = y + 3.5.$$

Read aloud: the conditional expectation of capital X given capital Y equals little y , equals little y plus three point five.

This means $E[X \mid Y]$ is itself a random variable, namely $Y + 3.5$. Taking expectations again recovers the unconditional mean (the *tower property*):

$$E[E[X \mid Y]] = E[Y + 3.5] = 3.5 + 3.5 = 7 = E[X].$$

Read aloud: expectation of expectation of capital X given capital Y equals expectation of capital Y plus three point five equals three point five plus three point five equals seven equals expectation of capital X .

The rest of the lesson develops these ideas in general.

1 Setup

Throughout, $(\Omega, \mathcal{F}, P)$ is a probability space and all random variables (RVs) are real-valued unless otherwise stated. We write $L^p = L^p(\Omega, \mathcal{F}, P)$ for the usual Lebesgue spaces, and for a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ we write $L^p(\mathcal{G})$ for the closed subspace of $\mathcal{G}$ -measurable representatives.

2 Conditional expectation given a random variable

Let X, Y be RVs with joint density $f_{X,Y}$ (the discrete case is analogous, replacing densities by PMFs and integrals by sums). Whenever $f_Y(y) > 0$, define the *conditional density of X given $Y = y$* by

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Read aloud: f sub capital X given capital Y of little x given little y equals f sub capital X comma capital Y of little x comma little y , divided by f sub capital Y of little y .

Definition 1 (Conditional expectation given $Y = y$). For each y with $f_Y(y) > 0$,

$$E[X \mid Y = y] = \int_{\mathbb{R}} x f_{X|Y}(x \mid y) dx,$$

Read aloud: expectation of capital X given capital Y equals little y , equals the integral over the real line of little x times f sub capital X given capital Y of little x given little y , d little x .

assuming the integral is well-defined. The map $g : y \mapsto E[X \mid Y = y]$ is a Borel function; the random variable

$$E[X \mid Y] := g(Y)$$

Read aloud: expectation of capital X given capital Y is defined as little g of capital Y .

is called the *conditional expectation of X given Y* .

3 Conditional expectation given a σ -algebra

Definition 2 (Conditional expectation given $\mathcal{G}$). Let $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub- σ -algebra. A random variable Z is a *version* of $E[X \mid \mathcal{G}]$ if

(i) Z is $\mathcal{G}$ -measurable;

(ii) $\int_A Z dP = \int_A X dP$ for every $A \in \mathcal{G}$.

Existence and a.s.-uniqueness of Z follow from the Radon–Nikodym theorem applied to the signed measure $\nu(A) := \int_A X dP$ on $\mathcal{G}$, which is absolutely continuous with respect to the restriction $P|_{\mathcal{G}}$. The earlier definition is recovered by taking $\mathcal{G} = \sigma(Y)$: by the Doob–Dynkin lemma, every $\sigma(Y)$ -measurable RV is of the form $g(Y)$, and one checks that $g(y) = E[X \mid Y = y]$ satisfies (i)–(ii).

4 The tower property

Theorem 3 (Tower property). Let $X \in L^1$ and let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be sub- σ -algebras. Then almost surely

$$E[E[X \mid \mathcal{G}] \mid \mathcal{H}] = E[X \mid \mathcal{H}].$$

Read aloud: expectation of, expectation of capital X given script G , given script H , equals expectation of capital X given script H .

In particular, taking $\mathcal{H} = \{\emptyset, \Omega\}$ gives the unconditional identity

$$E[E[X \mid \mathcal{G}]] = E[X].$$

Read aloud: expectation of expectation of capital X given script G , equals expectation of capital X .

Proof. Write $Z := E[X \mid \mathcal{G}]$ and let W be any version of $E[Z \mid \mathcal{H}]$; we show W is also a version of $E[X \mid \mathcal{H}]$.

(i) *$\mathcal{H}$ -measurability.* By the defining property of $E[Z \mid \mathcal{H}]$, the RV W is $\mathcal{H}$ -measurable.

(ii) *Defining integral identity.* Fix any $A \in \mathcal{H}$. Because $\mathcal{H} \subseteq \mathcal{G}$, we have $A \in \mathcal{G}$ as well. By the definition of $W = E[Z \mid \mathcal{H}]$,

$$\int_A W dP = \int_A Z dP. \quad (*)$$

Read aloud: the integral over capital A of capital W , d capital P , equals the integral over capital A of capital Z , d capital P ; star. By the definition of $Z = E[X \mid \mathcal{G}]$, applied to the set $A \in \mathcal{G}$,

$$\int_A Z dP = \int_A X dP. \quad (**)$$

Read aloud: the integral over capital A of capital Z, d capital P, equals the integral over capital A of capital X, d capital P; double star. Chaining (*) and (**) gives

$$\int_A W dP = \int_A X dP \quad \text{for every } A \in \mathcal{H}.$$

Read aloud: the integral over capital A of capital W, d capital P, equals the integral over capital A of capital X, d capital P, for every capital A in script H. Combined with (i), this is precisely the defining property of $E[X | \mathcal{H}]$. By a.s.-uniqueness of conditional expectations, $W = E[X | \mathcal{H}]$ a.s.

The unconditional form is the special case $\mathcal{H} = \{\emptyset, \Omega\}$, for which $E[\cdot | \mathcal{H}]$ collapses to the ordinary expectation. $\square$

Proposition 4 (L^2 projection). *If $X \in L^2$, then $E[X | \mathcal{G}]$ is the orthogonal projection of X onto $L^2(\mathcal{G})$. Equivalently,*

$$E[X | \mathcal{G}] = \arg \min_{Z \in L^2(\mathcal{G})} E[(X - Z)^2].$$

Read aloud: expectation of capital X given script G equals arg-min over capital Z in capital L two of script G, of expectation of, open paren capital X minus capital Z, close paren, squared.

Example 5 (Sum of two dice). Let Y and W be independent and uniform on $\{1, \dots, 6\}$, and let $X = Y + W$. Then for $y \in \{1, \dots, 6\}$,

$$E[X | Y = y] = y + E[W] = y + 3.5,$$

Read aloud: expectation of capital X given capital Y equals little y, equals little y plus expectation of capital W, equals little y plus three point five. so $E[X | Y] = Y + 3.5$. The tower property gives

$$E[E[X | Y]] = E[Y] + 3.5 = 3.5 + 3.5 = 7 = E[X]. \quad \square$$

Read aloud: expectation of expectation of capital X given capital Y, equals expectation of capital Y plus three point five, equals three point five plus three point five, equals seven, equals expectation of capital X.